

Question 1 (10 points)

Introduction

A Maclaurin series represents a function as an infinite sum of terms, which are computed from the values of the function's derivatives at zero. The Maclaurin series of the sine is:

$$\sin x \cong \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \quad (1)$$

In equation (1), both x and $\sin x$ are real numbers. They can be expressed as integer numbers to the detriment of precision. In particular, the argument of sine is multiplied by 10 (i.e., with substitution $y = 10x$) and then cast to an integer; the result is multiplied by 100 and cast to an integer. We obtain

$$100 \cdot \sin y \cong \sum_{n=0}^{\infty} \frac{100 \cdot (-1)^n (y)^{2n+1}}{(2n+1)! \cdot 10^{2n+1}} = 10y - \frac{y^3}{3! \cdot 10} + \frac{y^5}{5! \cdot 10^3} - \frac{y^7}{7! \cdot 10^5} + \dots \quad (2)$$

Assignment

Write a `Maclaurin` subroutine that computes the value of the Maclaurin series of the sine at a point x , up to order n . The subroutine receives in input the integer values y and n ; the returned value is 100 times the value of the Maclaurin series of $\sin y$.

Suggestion: according to equation (2), the element of order $n = 0$ is $t_0 = 10y$. Then, any new element t_i can be computed based on the previous element t_{i-1} as follows:

$$t_i = \frac{-t_{i-1} \cdot y^2}{(2i) \cdot (2i+1) \cdot 100} \quad (3)$$

Example: the input parameters are $y = 20$ and $n = 3$.

$$t_0 = 10y = 200$$

$$t_1 = (-200 \cdot 20^2) / (2 \cdot 3 \cdot 100) = -80000 / 600 = -133$$

$$t_2 = (133 \cdot 20^2) / (4 \cdot 5 \cdot 100) = 53200 / 2000 = 26$$

$$t_3 = (-26 \cdot 20^2) / (6 \cdot 7 \cdot 100) = -10400 / 4200 = -2$$

The returned value is $t_0 + t_1 + t_2 + t_3 = 200 - 133 + 26 - 2 = 91$.

Important: since the operation of division introduces an approximation, you can do it only after computing an element of the series. In other words, you have to compute the numerator and denominator of equation (3) first, then you divide the numerator by the denominator.

The suggested method guarantees the absence of overflow during computation (this is not true for other implementations).

Computer Architectures

Exam of 12/02/2025

A2

Question 2 (8 points)

Extend the previous exercise to play the A note when the user presses the button INT0.

In details:

- add proper initializations in `main()`
- in the handler of INT0, start timer 0 in order to generate an interrupt every $k = 1263$ clock cycles
- in `IRQ_timer.c`, define the global variable `int sineValues[45];`
- in the handler of timer 0, implement the following pseudocode:

```
static int repeat = 0;
static int ticks = 0;
int input, output;

if (repeat < 200)
{
    input = cast_to_int(1.428 * ticks);    // see Note 1 below
    output = 500 + Maclaurin(input, 3) / 2;
    sineValues[ticks + 22] = output;      // see Note 2 below
    write output to proper bits of D/A converter register
    ticks ++;
    if (ticks > 22)
    {
        ticks = -22;
        repeat += 1;
    }
}
else
    write zero to proper bits of D/A converter register
```

Note 1: for a better accuracy, before casting a float value to an integer value you should add 0.5 to the float value if it is positive. Instead, if the float value is negative, you should subtract 0.5.

Note 2: The values stored in the array `sineValues` are not used, but they must be present. They will be checked during the evaluation of the exam.

You do not need to handle consecutive presses of INT0: pressing it a second time has no effect. Therefore, button debouncing is not required for this project.